

In this page, I investigate some typical finite groups.

Specifically, investigate:

Abelian and Abelianization

Commutator and Derived subgroup

Upper-Central Series and Solvability and Composition Series

Center and Conjugacy classes.

Explicit all subgroup

The general properties of  $S_n$ ,  $D_n$ ,  $C_n$  will be covered in other page.

$\sqrt{D_{2p}} \times C_p$ , where  $p$  is an odd prime number.

$\sqrt{S_4}$ .

$$\begin{array}{l|l} S_4 = & \{ \text{id}, \\ 1,1,2 & (1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4), \\ 2,2 & (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3), \\ 1,3 & (1\ 2\ 3), (1\ 3\ 2), (1\ 3\ 4), (1\ 4\ 3), (1\ 2\ 4), (1\ 4\ 2), (2\ 3\ 4), (2\ 4\ 3), \\ 4 & (1\ 2\ 3\ 4), (1\ 2\ 4\ 3), (1\ 3\ 2\ 4), (1\ 3\ 4\ 2), (1\ 4\ 2\ 3), (1\ 4\ 3\ 2) \} \\ \text{Partition} & \end{array}$$

A4. Conjugacy classes.

$$A_4 = \{id, (12)(34), (13)(24), (14)(23), \\ (123), (132), (124), (142), (134), (143), \\ (234), (243)\}$$

Since the conjugate cannot change the cycle type, the conjugacy class of each element in  $A_4$  are contained the set of same cycle type. But we don't know it exactly equals to the set, let's calculate.

$$(123)(12)(34)(123)^{-1} = (23)(14)$$

$$(124)(12)(34)(124)^{-1} = (24)(31)$$

therefore, the conjugacy classes of  $(12)(34)$  equals the set of  $(2,2)$  cycles.

$$(132)(123)(132)^{-1} = (312) = (123) \quad (\text{In actually, } (123)^{-1} = (132))$$

$$(124)(123)(124)^{-1} = (243)$$

$$(142)(123)(142)^{-1} = (413) = (134)$$

$$(134)(123)(134)^{-1} = (324) = (243)$$

$$(143)(123)(143)^{-1} = (421) = (142)$$

$$(234)(123)(234)^{-1} = (134)$$

$$(243)(123)(243)^{-1} = (142)$$

$$(12)(34)(123)(12)(34)^{-1} = (214)$$

$$(13)(24)(123)(13)(24)^{-1} = (341) = (134)$$

$$(14)(23)(123)(14)(23)^{-1} = (432) = (243)$$

therefore, the conjugacy classes of  $(123)$  has just four elements.

Generally, since  $|G:G_\alpha| = C_\alpha$  and  $G_\alpha = C_G(\alpha)$  in the conjugate action, therefore number of elements of the orbit is  $|G|/|C_G(\alpha)|$ .

Since  $C_{A_4}((123)) = \{id, (123), (132)\},$

the orbit of  $(123)$  has  $|2/3| = 4$  elements

$A_5$ . The smallest group of non-abelian simple group.